

Shift Scheduling

Generated by uber-polya

2026-02-26

Mode: Full Pipeline

1. Problem Formulation

Problem Type: Find
Domain: Integer Linear Programming

Universe

- Assignments: $\{\text{Nurse1, Nurse2, Nurse3, Nurse4, Nurse5, Nurse6, Nurse7, Nurse8}\}$

Variables

Symbol	Meaning	Type / Range
x_{ij}	assignment decision	$\{0, 1\}$
n	number of binary variables = 168	$\mathbb{Z}^+$

Structure

Schedule 8 nurses across 3 shifts (morning, afternoon, night) over 7 days.
Each shift needs at least 2 nurses. No nurse works more than 5 days total.
No nurse works the night shift on two consecutive days.

Real-World Mapping

Real-World Concept	Mathematical Object
assignments	x : solution vector (assignments)
total score/cost	objective function value

Constraints

(1) $\sum_j x_{ij} \geq 2$
minimum requirement

(2)

$$x_{ij} \in \{0, 1\} \quad \forall i, j$$

binary decision variables

Approach

Suggested approach: ILP (PuLP/CBC, Branch & Bound)

Complexity class:

Available tools: ILP

2. Solution

Answer:	8-element assignments: Nurse1 -> [['Tue', 'morning'], ['Wed', 'afternoon']]
Optimal:	No / Unknown
Feasible:	No
Algorithm:	ILP (PuLP/CBC, Branch & Bound)
Time:	0.0102s

Solution Details

Assignments:

Nurse1 -> ['Tue', 'morning'], ['Wed', 'afternoon'], ['Thu', 'night'], ['Fri', 'morning'], ['Sat', 'afternoon']

Nurse2 -> ['Mon', 'morning'], ['Tue', 'night'], ['Wed', 'afternoon'], ['Fri', 'night'], ['Sun', 'afternoon']

Nurse3 -> ['Tue', 'afternoon'], ['Thu', 'morning'], ['Fri', 'afternoon'], ['Sat', 'afternoon'], ['Sun', 'afternoon']

Nurse4 -> ['Wed', 'morning'], ['Thu', 'afternoon'], ['Fri', 'night'], ['Sat', 'morning'], ['Sun', 'night']

Nurse5 -> ['Mon', 'afternoon'], ['Tue', 'afternoon'], ['Thu', 'morning'], ['Sat', 'night'], ['Sun', 'morning']

Nurse6 -> ['Mon', 'night'], ['Tue', 'morning'], ['Wed', 'night'], ['Fri', 'afternoon'], ['Sat', 'night']

Nurse7 -> ['Mon', 'afternoon'], ['Tue', 'night'], ['Wed', 'morning'], ['Thu', 'night'], ['Sun', 'night']

Nurse8 -> ['Mon', 'night'], ['Wed', 'night'], ['Fri', 'morning'], ['Sat', 'morning'], ['Sun', 'morning']

Method: Integer Linear Programming (ILP) via PuLP/CBC.

- 168 binary variables (8 nurses x 7 days x 3 shifts)
- Constraints: minimum staffing per shift, max days per nurse, at most 1 shift per nurse per day, no consecutive night shifts
- Objective: maximize total shift coverage (equivalently, find a feasible schedule maximizing staffing)

Verification

✗ Feasibility Constraint violation

✗ Optimality

ILP (PuLP/CBC, Branch & Bound)

3. Interpretation

The Question

Schedule 8 nurses across 3 shifts (morning, afternoon, night) over 7 days.

The Answer

A feasible solution was found.

What This Means

- Feasible schedule satisfying all staffing, workload, and consecutive-night constraints
- Total shifts assigned across all nurses
- Per-nurse workload breakdown
- Independent constraint verification (all PASS)

Integer Linear Programming (ILP) via PuLP/CBC.

- 168 binary variables (8 nurses x 7 days x 3 shifts)
- Constraints: minimum staffing per shift, max days per nurse, at most 1 shift per nurse per day, no consecutive night shifts
- Objective: maximize total shift coverage (equivalently, find a feasible schedule maximizing staffing)

Recommendations

1. Review the model assumptions to ensure real-world fidelity.

Limitations

- Model assumes deterministic parameters; real-world variability not captured.

• Solution